

Sylow Theorems proofs

1st proof: let F be the set of all subsets of G of order p^e , one of those subsets is the one we are looking for $\rightarrow$ show it has a stabilizer of order p^e .
 Lemma: number of subsets of order p^e in a set of $n = p^e \cdot m$ elements, $p \nmid m$, is the binomial coefficient $N = \binom{n}{p^e} = \frac{n(n-1)\dots(n-p^e+1)}{p^e(p^e-1)\dots 1}$, moreover $p \nmid N$ for each $\frac{p^e m - k}{p^e - k}$, and p^a the highest power of p that divides k with $p^j < p^e$ if $k > 0$ p 's in the top cancel out with p 's in the bottom, if $k = 0$: $\frac{p^e m}{p^e} = m$ so $p \nmid N$.
 decompose F into orbits for the operation of left multiplication
 $N = |F| = \sum_{\text{orbits}} |O|$, since $p \nmid N$, some orbit has an order not divisible by p , because if all were divisible by p then so would be N , say the orbit of the subset $U: pX|O_U|$, for $|stab(U)| \rightarrow$ elements in a single block, and, $|O_U| \rightarrow$ total number of blocks: $|stab(U)| \cdot |O_U| = |G| = p^e m \Rightarrow |stab(U)| = p^e$ this stabilizer is the required subgroup.

2nd proof: subgroup K and Sylow p -subgroup H of G , $S = G/H$ set of all cosets, K is a p -group, $p \mid |K|$, G acts transitively on S so S forms a single orbit $O_S = S$, $\forall s \in S$ (two orbits are either the same or completely different)
 H is the stabilizer $G_s = \{g \in G \mid gs = s\} = H$ for some $s \in S$ namely of $s = 1 \cdot H$, $G_{1 \cdot H} = H$ for any $as = aH$ we look for g such that $g(aH) = aH \Rightarrow a^{-1}gaH = H \Rightarrow a^{-1}ga \in H = g \in aHa^{-1}$ so $G_{as} = G_{aH} = aHa^{-1}$, let's call that $H' = aHa^{-1}$ which is the stabilizer of any random block $aH \in S = G/H$, restrict G to K , so only $k \in K$ can act, because K has fewer elements it might not act transitively on S so the one orbit shatters to independent K -orbits $|S| = [G:H]$ and $p \nmid |S|$ but $|S| = \text{sum of sizes of } K\text{-orbits}$, there's at least one orbit whose order isn't divisible by p , say that's O , for $H' = aHa^{-1}$ the stabilizer of as with orbit O for the operation of G , then the stabilizer of as in the restricted operation of $K \subset G$ is obviously $K \cap H'$ and $|O| = [K : H' \cap K]$ prime to p , because $|orbit| = \frac{\text{size of acting group}}{\text{size of stabilizer}}$, ex $|O_S| = |G|/|G_S|$ but H' is a conjugate of H so it's also a p -group and $H' \cap K$ is a p -group, because $H' \cap K$ is a subgroup of H'
 We proved that the index of $H' \cap K$ isn't divisible by p , so it's a Sylow p -subgroup in K .
 3rd proof: All Sylow subgroups of G are conjugate to one say H , so the number of Sylow subgroups is $S = [G : N(H)]$, $N(H) = G_H$ under conjugation and $S = |O_H|$ because $|O_H| = \frac{|G|}{|G_H|} = \frac{|G|}{|N(H)|} \Rightarrow S = [G : N(H)]$

since $H \leq N$, $[G:N]$ divides $[G:H] = m$ because $[G:H] = [G:N] \cdot [N:H]$
 so $m = s [N:H]$ and we proved that $s \mid m$

Decompose set of Sylow subgroups $\{H_1, \dots, H_s\}$ to orbits for the operation of conjugation by $H = H_1$, so we scatter the one orbit to independent H -orbits
 the size of every orbit must divide the size of the group doing the acting
 the order of H is a power of p and so are $|O_H|$ smaller powers of p , $1, p, p^2, \dots$, an orbit consists of a single group H_i if and only if H is contained in $N_i = N(H_i)$ and so $H = H_i$ because $H_i \triangleleft N_i$ and H_i is a Sylow p -subgroup so it's the only one, because H_i is a Sylow p -subgroup of G .
 Therefore, there's only one orbit with one element, $\{H\} = O_H \rightarrow |O_H| = 1$.
 All others have order divisible by p , because their orders divide $|H|$.
 $s = 1 + (\text{multiple of } p) + (\text{multiple of } p) + \dots = 1 + k \cdot p \Rightarrow s \equiv 1 \pmod{p}$